

# Hein Zarni Aung

heinzarniaung.dev@gmail.com | linkedin.com/in/heaung | github.com/LoistCloist

## EDUCATION

---

**Stony Brook University** — Expected Graduation: May 2027

*B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40*

*Honors & Scholarships: Global Excellence Scholarship*

*Relevant Coursework: Statistics & Probability, Algorithms, Systems Fundamentals I & II, Full Stack Development*

## RESEARCH

---

**Undergraduate Research Assistant — Ethos Lab, SBU** — Sep 2025 – Dec 2025 (paused)

*PI: Abisheka Pitumpe | Team size: 2*

- Built a real-time data pipeline ingesting and parsing newly issued SSL/TLS certificate and site data (HTML, CSS, JS, URLs) to flag fraudulent domains impersonating EZ-Pass tolling services.
- Used Selenium to automate certificate and domain scraping and Python to filter and triage flagged sites; auto-captured homepage screenshots of promising candidates to speed up manual review.
- Collaborated in a two-person team with one honors student under a faculty PI to design keyword- and heuristic-based detection logic for scam website identification.

## INTERNSHIPS

---

**Software Engineering Intern — Clustr** — Jan 2026 – Sept 2026 (expected)

*Cross-platform React Native social/scheduling app | ~8-person dev team*

- Built server-side APIs using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced permission checks, join/leave workflows, and activity feed endpoints.
- Contributed 50+ commits and implemented authentication features (password reset flow, custom claims), regularly resolving conflicts across feature branches with an 8-person team via Git.

**Software Engineering Intern — Prenora Dynamics LLC** — Jun 2026 – Sept 2026 (expected)

*~8-person team*

- Researched and evaluated third-party SDKs and APIs to inform build-vs-buy decisions for an in-house video chat feature, working across client and backend systems end-to-end.

## PROJECTS

---

**TuneAcademy — Hack Brooklyn 2026 (36-hour hackathon)** — Apr 2026

*Winner: Best Prenora Dynamics Hack (Sponsor Challenge) | Team of 4*

- Built a data-processing pipeline extracting pitch, rhythm, and tone features from raw audio recordings using Essentia/NumPy, transforming unstructured signal data into structured inputs for a downstream matching tool.
- Pivoted from a planned machine learning model to rule-based scoring algorithms under hackathon time and compute constraints, delivering reliable weakness detection within the 36-hour build window.
- Collaborated in a 4-person team; broader app included a React/Tailwind frontend, Python/FastAPI backend in Docker, Cloud Firestore, and GCP hosting.

**Discord Voice Channel Activity Monitor** — Node.js, Discord.js, Cloud

- Built a Node.js/Discord.js bot providing real-time monitoring and notifications when users join a voice channel, with customizable DM alerts and per-server/per-user thresholds via JSON-based configuration.
- Deployed as an operational service on a cloud server using PM2 for reliability; reached 50 concurrent active users.

## SKILLS

---

**Languages:** Python, SQL, TypeScript, Java, C

**Backend:** Node.js, Express.js, FastAPI

**Databases & Cloud:** PostgreSQL (SQL), MongoDB, Firebase/Firestore, AWS, GCP

**Tools:** Git, Docker, Linux, PM2, Selenium

**Frontend:** React, React Native, Next.js